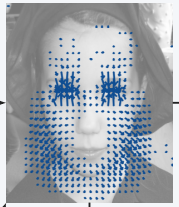
One commercial render, split into single operations with exact footprints



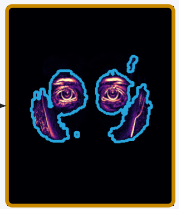
original O



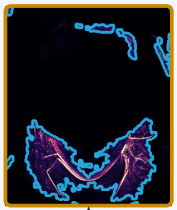
optical flow f
$O(x+f) \approx R(x)$



eye mask m_e
from landmarks



eye
$O(x + m_e f)$



contour
$O(x + (1-m_e)f)$



render R



residual
$P = R - O(x+f)$

high = P − low

Gaussian low-pass
σ = 6 px



tone
O + low(P)



texture
O + high(P)

What it gives

Exact
$O(x+f) + P = R$ by construction;
108 dB
reconstruction

A footprint each
$M = |C - O| > 6$ of 255, dilated (cyan)

Yield
50,112
single-operation items from 18,734 renders (Tencent, Megvii); Alibaba is never decomposed